

CHAPTER 2

Motion in a Straight Line

Distance, Displacement, Speed and Velocity

1. A particle is moving along x -axis with its position (x) varying with time (t) as $x = \alpha t^4 + \beta t^2 + \gamma t + \delta$. The ratio of its initial velocity to its initial acceleration, respectively, is: (2024 Re)
 - a. $2\alpha : \delta$
 - b. $\gamma : 2\delta$
 - c. $4\alpha : \beta$
 - d. $\gamma : 2\beta$
2. A vehicle travels half the distance with speed v and the remaining distance with speed $2v$. Its average speed is: (2023)

- a. $\frac{3v}{4}$
- b. $\frac{v}{3}$
- c. $\frac{2v}{3}$
- d. $\frac{4v}{3}$

3. The position of a particle is given by

$$\vec{r}(t) = 4t\hat{i} + 2t^2\hat{j} - 5\hat{k}$$

where t is in seconds and r in metre. Find the magnitude and direction of velocity $v(t)$, at $t = 1$ s, with respect to x -axis (2023-Manipur)

- a. $4\sqrt{2} \text{ ms}^{-1}, 45^\circ$
- b. $4\sqrt{2} \text{ ms}^{-1}, 60^\circ$
- c. $3\sqrt{2} \text{ ms}^{-1}, 30^\circ$
- d. $3\sqrt{2} \text{ ms}^{-1}, 45^\circ$

4. If $\vec{F} = 2\hat{i} + \hat{j} - \hat{k}$ and $\vec{r} = 3\hat{i} + 2\hat{j} - 2\hat{k}$, then the scalar

and vector products of $\vec{F}$ and $\vec{r}$ have the magnitudes respectively as: (2022 Re)

- a. 10, 2
- b. $3\sqrt{3}$
- c. $4\sqrt{5}$
- d. $10\sqrt{2}$

5. Preeti reached the metro station and found that the escalator was not working. She walked up the stationary escalator in time t_1 . On other days, if she remains stationary on the moving escalator, then the escalator takes her up in time t_2 . The time taken by her to walk up on the moving escalator will be: (2017-Delhi)

- a. $\frac{t_1 t_2}{t_2 - t_1}$
- b. $\frac{t_1 t_2}{t_2 + t_1}$
- c. $t_2 - t_1$
- d. $\frac{t_1 + t_2}{2}$

6. Two cars P and Q start from a point at the same time in a straight line and their positions are represented by $X_P(t) = at + bt^2$ and $X_Q(t) = ft - t^2$. At what time do the cars have the same velocity? (2016 - II)

- a. $\frac{a+f}{2(1+b)}$
- b. $\frac{f-a}{2(1+b)}$
- c. $\frac{a-f}{1+b}$
- d. $\frac{a+f}{2(b-1)}$

7. If the velocity of a particle is $v = At + Bt^2$, where A and B are constants, then the distance travelled by it between 1 s and 2 s is: (2016 - I)

- a. $\frac{3}{2}A + 4B$
- b. $3A + 7B$
- c. $\frac{3}{2}A + \frac{7}{3}B$
- d. $\frac{A}{2} + \frac{B}{3}$

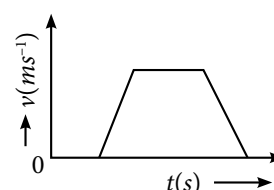
Acceleration

8. A particle of unit mass undergoes one dimensional motion such that its velocity varies according to $v(x) = \beta x^{-2n}$ where β and n are constants and x is the position of the particle. The acceleration of the particle as a function of x , is given by: (2015)

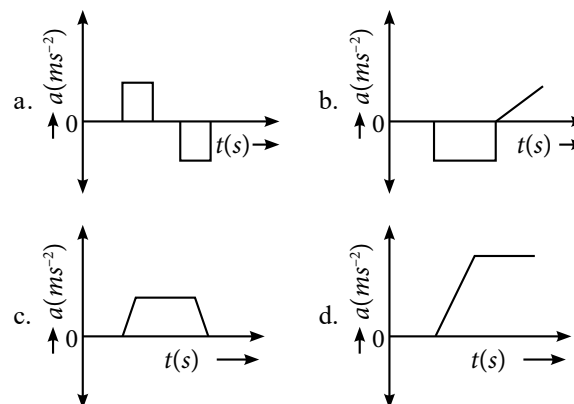
- a. $-2n\beta^2 x^{-4n-1}$
- b. $-2\beta^2 x^{-2n+1}$
- c. $-2n\beta^2 e^{-4n+1}$
- d. $-2n\beta^2 x^{-2n-1}$

Graphs

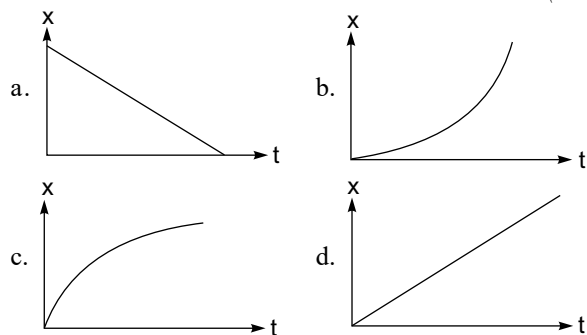
9. The velocity (v) – time (t) plot of the motion of a body is shown below: (2024)



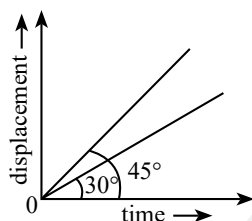
The acceleration (a) – time (t) graph that best suits this motion is:



10. The position-time ($x-t$) graph for positive acceleration is: (2022 Re)



11. The displacement time graphs of two moving particle make angles of 30° and 45° with the x-axis as shown in the figure. The ratio of their respective velocity is : (2022)



- a. $1:\sqrt{3}$ b. $\sqrt{3}:1$ c. $1:1$ d. $1:2$

Motion under Gravity

12. The ratio of the distance traveled by a freely falling body in the 1st, 2nd, 3rd and 4th second : (2022)
a. $1:1:1:1$ b. $1:2:3:4$
c. $1:4:9:16$ d. $1:3:5:7$
13. A ball is thrown vertically downward with a velocity of 20 m/s from the top of a tower. It hits the ground after some time with a velocity of 80m/s. The height of the tower is : ($g = 10 \text{ m/s}^2$) (2020)
a. 340 m b. 320 m c. 300 m d. 360 m
14. A person sitting in the ground floor of a building notices through the window, of height 1.5 m, a ball dropped from the roof of the building crosses the window in 0.1 s. What is the velocity of the ball when it is at the **topmost point** of the window? ($g = 10 \text{ m/s}^2$) (2020-Covid)
a. 14.5 m/s b. 4.5 m/s c. 20 m/s d. 15.5 m/s
15. A stone falls freely under gravity. It covers distances h_1 , h_2 and h_3 in the first 5 seconds, the next 5 seconds and the next 5 seconds respectively. The relation between h_1 , h_2 and h_3 is: (2013)
a. $h_1 = h_2 = h_3$ b. $h_1 = 2h_2 = 3h_3$
c. $h_1 = \frac{h_2}{3} = \frac{h_3}{5}$ d. $h_2 = 3h_1$ and $h_3 = 3h_2$

Answer Key

1. (d) 2. (d) 3. (a) 4. (d) 5. (b) 6. (b) 7. (c) 8. (a) 9. (a) 10. (b)
11. (a) 12. (d) 13. (c) 14. (a) 15. (c)